

AFFINE CONNECTIONS, CURVATURE, AND TORSION

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1. Conventions

1.1. Asymptotic notation

Let A be a topological space, $f: A \times \mathbb{R}^n \rightarrow \mathbb{R}$, and $g: \mathbb{R}^n \rightarrow [0, \infty)$. For a point $x_0 \in \mathbb{R}^n$, we say that

$$f(\alpha, x) = O_\alpha(g(x)) \text{ as } x \rightarrow x_0 \quad (1.1)$$

if there exists a neighborhood U of x_0 and a continuous function $C: \mathbb{R}^n \rightarrow [0, \infty)$ such that $|f(\alpha, x)| \leq C(\alpha)g(x)$ for all points $x \in U$ and $\alpha \in A$.

2. The piecewise-smooth Moore path category

Definition 2.1. Let M be a smooth manifold. The *piecewise-smooth Moore path category* $P(M)$ is the category defined as follows:

1. Objects of $P(M)$ are given by points $p \in M$.
2. Given objects $x, y \in P(M)$, a morphism $\gamma: x \rightsquigarrow y$ is given by a piecewise-smooth path $\gamma: [0, t] \rightarrow M$ of duration $t(\gamma) \geq 0$ such that $\gamma(0) = x$ and $\gamma(t) = y$.
3. Given objects $x, y, z \in P(M)$ and morphisms $\alpha: y \rightsquigarrow z$ and $\beta: x \rightsquigarrow y$, their composition is the piecewise-smooth path $\alpha * \beta: x \rightsquigarrow z$ of duration $t(\alpha) + t(\beta)$ defined by

$$(\beta * \alpha)(t) = \begin{cases} \alpha(t) & \text{if } t \leq t(\alpha) \\ \beta(t - t(\alpha)) & \text{if } t > t(\alpha) \end{cases}. \quad (2.1)$$

At any point $x \in P(M)$, the identity morphism $\text{id}_x: x \rightsquigarrow x$ is the constant path of duration 0 at x . Besides this morphism, no other morphisms in this category are invertible; however, we may define for each path $\gamma: x \rightsquigarrow y$ the *reverse* morphism

$$\bar{\gamma}: y \rightsquigarrow x; \quad \bar{\gamma}(t) = \gamma(t(\gamma) - t) \quad (2.2)$$

of the same duration. Taking reverses yields an isomorphism of categories $P(M) \cong P(M)^{\text{op}}$. Also, if M and N are smooth manifolds and $f: M \rightarrow N$ is a smooth map, then f may be regarded as a functor $f: P(M) \rightarrow P(N)$ between path categories whose action on paths is given by $f(\gamma) = f \circ \gamma$.

Definition 2.2. Let E be an affine space. Given points $p, q \in E$, define the straight-line path $[p \rightarrow q]: p \rightsquigarrow q$ of duration 1 by

$$[p \rightarrow q](t) = (q - p) \cdot t + p. \quad (2.3)$$

Definition 2.3. Let E be an affine space associated to a vector space V . Given $p \in E$ and $v, w \in V$, define the *parallelogram path* $\llbracket v, w \rrbracket_p$ based at p to be the path

$$\llbracket v, w \rrbracket_p = [(v \rightarrow 0) * (v + w \rightarrow v) * (w \rightarrow v + w) * (0 \rightarrow w)] + p. \quad (2.4)$$

3. Torsion

Theorem 3.1. Let ∇ be an affine connection on $\mathbb{R}^n$, and

Proposition 3.2. Let M be a smooth manifold with affine connection ∇ . If γ is a piecewise-smooth path, then

$$\tau_{\nabla}(\bar{\gamma}) = -\Gamma(\gamma)\tau_{\nabla}(\gamma). \quad (3.1)$$

Proof. The accumulated torsion of the reverse path γ is given by

$$\tau_{\nabla}(\bar{\gamma}) = \int_0^{t(\bar{\gamma})} \Gamma(\bar{\gamma})_t^0 [\bar{\gamma}'(t)] dt = - \int_0^{t(\gamma)} \Gamma(\bar{\gamma})_{t(\bar{\gamma})}^0 \Gamma(\bar{\gamma})_t^{t(\bar{\gamma})} [\gamma'(t(\gamma) - t)] dt. \quad (3.2)$$

Performing the substitution $t' = t(\gamma) - t$ and applying Proposition ?? yields

$$\tau_{\nabla}(\bar{\gamma}) = -\Gamma(\gamma) \int_0^{t(\gamma)} \Gamma(\gamma)_{t'}^0 [\gamma'(t')] dt'. \quad (3.3)$$

The conclusion follows. ■

Proposition 3.3. Let M be a smooth manifold with affine connection ∇ . If $\alpha: y \rightsquigarrow z$ and $\beta: x \rightsquigarrow y$ are smooth paths, then

$$\tau_{\nabla}(\alpha\beta) = \Gamma(\bar{\beta})[\tau_{\nabla}(\alpha)] + \tau_{\nabla}(\beta). \quad (3.4)$$

Proof. The endpoint of the development of $\alpha\beta$ is given by

$$\tau_{\nabla}(\alpha\beta) = \int_{t(\beta)}^{t(\alpha)+t(\beta)} \Gamma(\alpha\beta)_t^0 [(\alpha\beta)'(t)] dt + \int_0^{t(\beta)} \Gamma(\alpha\beta)_t^0 [(\alpha\beta)'(t)] dt. \quad (3.5)$$

Applying Proposition ?? yields

$$\tau_{\nabla}(\alpha\beta) = \int_0^{t(\alpha)} \Gamma(\bar{\beta})\Gamma(\alpha)_t^0[\alpha'(t)] dt + \int_0^{t(\beta)} \Gamma(\beta)_t^0[\beta'(t)] dt \quad (3.6)$$

$$= \Gamma(\bar{\beta}) \int_0^{t(\alpha)} \Gamma(\alpha)_t^0[\alpha'(t)] dt + \int_0^{t(\beta)} \Gamma(\beta)_t^0[\beta'(t)] dt. \quad (3.7)$$

The conclusion follows. ■

Note that this implies a kind of “right cancellation” law for τ_{∇} : $\tau_{\nabla}(\alpha\beta) = \tau_{\nabla}(\alpha'\beta)$ if and only if $\tau_{\nabla}(\alpha) = \tau_{\nabla}(\alpha')$. Left cancellation does not immediately hold, due to the prescence of the parallel transport term. However, if β is a loop such that $\Gamma(\beta)$ is the identity, then the torsion satisfies $\tau_{\nabla}(\alpha\beta) = \tau_{\nabla}(\alpha) + \tau_{\nabla}(\beta)$. In this sense, the accumulated torsion around loops is “additive” when a connection has trivial holonomy.

Proposition 3.4. *Let M be a smooth manifold with affine connection ∇ . If $\alpha_1, \alpha_2: x \rightsquigarrow y$ are piecewise smooth paths and $\gamma: x \rightsquigarrow y$ is another such path, then*

$$\tau_{\nabla}(\bar{\alpha}_2\alpha_1) = \tau_{\nabla}(\bar{\alpha}_2\gamma \cdot \bar{\gamma}\alpha_1). \quad (3.8)$$

Proof. By Proposition ??, it suffies to show that $\tau_{\nabla}(\bar{\alpha}_2) = \tau_{\nabla}(\bar{\alpha}_2 \cdot \gamma\bar{\gamma})$. But $\gamma\bar{\gamma}$ has trivial parallel transport, so by Proposition ?? again we have

$$\tau_{\nabla}(\bar{\alpha}_2 \cdot \gamma\bar{\gamma}) = \tau_{\nabla}(\bar{\alpha}_2) + \tau_{\nabla}(\gamma\bar{\gamma}). \quad (3.9)$$

Finally, by Proposition ?? and ??, we have

$$\tau_{\nabla}(\gamma\bar{\gamma}) = \Gamma(\gamma)[\tau_{\nabla}(\gamma)] + \tau_{\nabla}(\bar{\gamma}) = \Gamma(\gamma)[\tau_{\nabla}(\gamma)] - \Gamma(\gamma)[\tau_{\nabla}(\gamma)] = 0.$$

The conclusion follows. ■

By using the laws we have just established, we may express the accumulated torsion over a contractible loop in terms of an integral of the infinitesimal torsion within the region bounded by the loop. We adopt the following notation to describe paths in an interval $I \subseteq \mathbb{R}$: given $a, b \in I$, define $a \rightarrow b$ to be the straight-line path

$$[a \rightarrow b]: a \rightsquigarrow b; \quad [a \rightarrow b](t) = (1-t) \cdot a + t \cdot b \quad (3.10)$$

from a to b .

Theorem 3.5. *Let M be a smooth manifold with flat affine connection ∇ . If $\gamma: x \rightsquigarrow x$ is contractible via the homotopy¹ $H: [0, 1] \times [0, t(\gamma)] \rightarrow M$ with $H_1 = \gamma$, then*

$$\tau_{\nabla}(\gamma) = \iint_{[0,1] \times [0,t(\gamma)]} \Gamma H(s \rightarrow 0, t) \left[T_{\nabla} \left(\frac{\partial H}{\partial s}, \frac{\partial H}{\partial t} \right) \right] ds dt \quad (3.11)$$

Proof. Given $s \in [0, 1]$ or $t \in [0, t(\gamma)]$, we view the maps $H(s, -)$ and $H(-, t)$ as functors between path categories; when applied to straight-line paths in their respective domains, these yield “vertical” or “horizontal” paths in M . Let $n > 0$ be an integer. By Proposition ??, the fact that the homotopy fixes endpoints, and Proposition ??, the torsion accumulated over $\gamma = H_1$ is equal to

$$\tau_{\nabla}(H(1, 0 \rightarrow 1)) = \tau_{\nabla} \left[\prod_{k=0}^{n-1} H(1, \frac{k}{n} \rightarrow \frac{k+1}{n}) \right] \quad (3.12)$$

$$= \tau_{\nabla} \left[\prod_{k=0}^{n-1} H\left(1 \rightarrow 0, \frac{k+1}{n}\right) \cdot H\left(1, \frac{k}{n} \rightarrow \frac{k+1}{n}\right) \cdot H\left(0 \rightarrow 1, \frac{k}{n}\right) \right] \quad (3.13)$$

$$= \sum_{k=0}^{n-1} \tau_{\nabla} \left[H\left(1 \rightarrow 0, \frac{k+1}{n}\right) \cdot H\left(1, \frac{k}{n} \rightarrow \frac{k+1}{n}\right) \cdot H\left(0 \rightarrow 1, \frac{k}{n}\right) \right]. \quad (3.14)$$

Each “wedge” in the sum ?? may be broken into individual rectangular units as follows. For integers $j \in \llbracket 1, n \rrbracket$ and $k \in \llbracket 0, n-1 \rrbracket$, define “subwedges” $\omega_{j,k}: x \rightsquigarrow x$ by

$$\omega_{j,k} = \underbrace{H\left(\frac{j}{n} \rightarrow 0, \frac{k+1}{n}\right) \cdot H\left(\frac{j}{n}, \frac{k}{n} \rightarrow \frac{k+1}{n}\right)}_{\alpha_{j,k}} \cdot \underbrace{H\left(0 \rightarrow \frac{j}{n}, \frac{k}{n}\right)}_{\beta_{j,k}}, \quad (3.15)$$

and “cells” $\rho_{j,k}$ by

$$\rho_{j,k} = H\left(\frac{j-1}{n}, \frac{k+1}{n} \rightarrow \frac{k}{n}\right) \cdot H\left(\frac{j}{n} \rightarrow \frac{j-1}{n}, \frac{k+1}{n}\right) \cdot H\left(\frac{j}{n}, \frac{k}{n} \rightarrow \frac{k+1}{n}\right) \cdot H\left(\frac{j-1}{n} \rightarrow \frac{j}{n}, \frac{k}{n}\right). \quad (3.16)$$

¹ooga booga

Then by Proposition ??, the accumulated torsion is

$$\tau_{\nabla}(\omega_{j,k}) = \Gamma(\bar{\beta}_{j-1,k})[\tau_{\nabla}(\alpha_{j-1,k} \cdot \rho_{j,k})] + \tau_{\nabla}(\beta_{j-1,k}) \quad (3.17)$$

$$= \Gamma(\bar{\beta}_{j-1,k})[\tau_{\nabla}(\alpha_{j-1,k}) + \tau_{\nabla}(\rho_{j,k})] + \tau_{\nabla}(\beta_{j-1,k}) \quad (3.18)$$

$$= \underbrace{\left[\Gamma(\bar{\beta}_{j-1,k})[\tau_{\nabla}(\alpha_{j-1,k})] + \tau_{\nabla}(\beta_{j-1,k}) \right]}_{\tau_{\nabla}(\omega_{j-1,k})} + \Gamma(\bar{\beta}_{j-1,k})[\tau_{\nabla}(\rho_{j,k})] \quad (3.19)$$

It follows that

$$\tau_{\nabla}(\gamma) = \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} \left[\Gamma H\left(\frac{j}{n} \rightarrow 0, \frac{k}{n}\right) \right] \tau_{\nabla}(\rho_{j,k}). \quad (3.20)$$

By Theorem ??, one recovers the desired expression as $n \rightarrow \infty$. ■

Lemma 3.6. *Let M be a smooth manifold with affine connection ∇ . If $h: \mathbb{R} \rightarrow M$ is a smooth curve, then*

$$e \quad (3.21)$$

Proof. Let $p \in \mathbb{R}$. There exists an open set $U \subseteq M$ containing $h(p)$ on which there exists a moving frame $E_1, \dots, E_n: U \rightarrow TM|_U$. In this frame, the connection is expressible in terms of a linear operator. Explicitly, let $I = h^{-1}(U)$ and $\gamma = h|_I$, and suppose that $V: I \rightarrow \gamma^*(TM|_U) \cong I \times \mathbb{R}^n$ is a vector field along γ . Then

$$(\gamma^* \nabla)_{d/dt} V = \frac{dV}{dt} + \omega V \quad (3.22)$$

where $\omega(t): T_{\gamma(t)}M \rightarrow T_{\gamma(t)}M$ is the endomorphism defined by $\omega E_i = \nabla_{d/dt} E_i$. ■

Lemma 3.7. *Let $U \subseteq \mathbb{R}^n$ be open, ∇ be an affine connection on U , and A be a smooth manifold. For a smooth family of smooth curves $\{\gamma_\alpha : I \rightarrow U\}_{\alpha \in A}$, we have*

$$\tau_\nabla[\gamma_\alpha(0 \rightarrow t)] = t \cdot \gamma'_\alpha(0) + O_\alpha(t^2). \quad (3.23)$$

Proof. For each $\alpha \in A$, let $\omega^\alpha = \omega_1^\alpha dt \in \Omega^1(I, \mathfrak{gl}_n)$ be the connection 1-form associated to $\gamma_\alpha^* \nabla$. Let $J \subseteq I$ be some bounded interval containing 0. By Taylor's theorem, for all $t \in J$, we have

$$\Gamma(\gamma_\alpha)_t^0[\gamma'_\alpha(t)] = [\mathbf{1} + \mathcal{E}_\alpha(t)]\gamma'_\alpha(t), \quad (3.24)$$

where $\|\mathcal{E}_\alpha(t)\|_{\max} \leq \sup_J \|\omega_1^\alpha\|_{\max} \cdot t$. Integrating, we have for all $t \in J$,

$$\tau_\nabla[\gamma_\alpha(0 \rightarrow t)] = [\gamma_\alpha(t) - \gamma_\alpha(0)] + \epsilon_\alpha(t), \quad (3.25)$$

where

$$|\epsilon_\alpha(t)|_{\max} \leq \int_0^1 |\mathcal{E}_\alpha(t)\gamma'_\alpha(t)|_{\max} dt \leq n \cdot \sup_J \|\omega_1^\alpha\|_{\max} \cdot \sup_J |\gamma'_\alpha| \cdot \frac{1}{2}t^2. \quad (3.26)$$

Since ω_1 and γ' are smooth in α , both supremums are continuous in α , so $\epsilon_\alpha(t) = O_\alpha(t^2)$. Applying Taylor's theorem again, for all $t \in J$ we may write

$$\gamma_\alpha(t) - \gamma_\alpha(0) = t \cdot \gamma'_\alpha(0) + \delta_\alpha(t), \quad (3.27)$$

where $|\delta_\alpha(t)|_{\max} \leq \sup_J |\gamma''_\alpha| \cdot \frac{1}{2}t^2$. Since γ'' is smooth in α , it follows that $\delta_\alpha(t) = O_\alpha(t^2)$ as well. The conclusion follows. $\blacksquare$

Theorem 3.8. *Let $U \subseteq \mathbb{R}^n$ be open and ∇ be an affine connection on U . Then for all points $p \in U$ and vectors $u, v \in \mathbb{R}^n$, we have*

$$\lim_{x, y \rightarrow 0} \frac{1}{xy} \cdot \tau_\nabla[x \cdot u, y \cdot v]_p = T_p(u, v). \quad (3.28)$$

Proof. For brevity, we may write $[x \cdot u, y \cdot v]_p = \gamma_4 * \gamma_3 * \gamma_2 * \gamma_1$, where γ_i is the i th side of the parallelogram. Then by Proposition ??, we have

$$\tau_\nabla[x \cdot u, y \cdot v]_p = \sum_{k \leq 4} \left[\prod_{i < k} \Gamma(\overline{\gamma_i}(x, y)) \right] \tau_\nabla(\gamma_k(x, y)). \quad (3.29)$$

The accumulated torsion for each smooth family of curves is given by

$$\tau_\nabla(\gamma_{x,y}^1(0 \rightarrow t)) = +t \cdot u + O_{x,y}(t^2) \quad \tau_\nabla(\gamma_{x,y}^2(0 \rightarrow t)) = +t \cdot v + O_{x,y}(t^2) \quad (3.30)$$

$$\tau_\nabla(\gamma_{x,y}^3(0 \rightarrow t)) = -t \cdot u + O_{x,y}(t^2) \quad \tau_\nabla(\gamma_{x,y}^4(0 \rightarrow t)) = -t \cdot v + O_{x,y}(t^2) \quad (3.31)$$

Likewise, the parallel transport along each family of curves is given by

$$\Gamma(\gamma_{x,y}^i(t \rightarrow 0)) = \mathbf{1} + t \cdot \omega_{p_{x,y}^i}(e_i) + O_{x,y}(t^2) \quad (3.32)$$

as $t \rightarrow 0$. Therefore, for $1 \leq k \leq 4$, we may write

$$\left[\prod_{i < k} \Gamma(\gamma_{x,y}^i(t_i \rightarrow 0)) \right] \tau_{\nabla}(\gamma_{x,y}^k(0 \rightarrow t_k)) = t_k e_k + \sum_{i < k} t_k t_i [\omega_{p_{x,y}^i}(e_i)] e_k + \mathcal{E}_{x,y}^k(\mathbf{t}) \quad (3.33)$$

where $\mathcal{E}^k: A \times I^k \rightarrow \mathbb{R}^n$ is $\sum_{i < k} O_{x,y}(t_i^2) + \sum_{i,j < k} O_{x,y}(t_i t_j t_k)$ as $\mathbf{t} \rightarrow 0$. Then

$$\sum_{k \leq 4} \left[\prod_{i < k} \Gamma(\gamma_{x,y}^i(t_i \rightarrow 0)) \right] \tau_{\nabla}(\gamma_{x,y}^k(0 \rightarrow t_k)) = \sum_{k \leq 4} t_k e_k + \sum_{k \leq 4} \sum_{i < k} t_k t_i [\omega_{p_{x,y}^i}(e_i)] e_k + \mathcal{E}_{x,y}(\mathbf{t}), \quad (3.34)$$

where $\mathcal{E}: A \times I^4 \rightarrow \mathbb{R}^n$ is the sum of the individual error terms $\mathcal{E}^k$. Thus, there exists a neighborhood U of the origin $\mathbf{0} \in I^4$ and a continuous map $K: A \rightarrow \mathbb{R}$ such that for all $\mathbf{t} \in U$, we have

$$|\mathcal{E}_{x,y}(\mathbf{t})|_{\max} \leq \frac{1}{4} K(x, y) \sum_{k \leq 4} \sum_{i,j < k} |t_i t_j t_k| \leq K(x, y) \sum_{i,j,k} |t_i t_j t_k|. \quad (3.35)$$

Now, to connect each path, we take $\mathbf{t} = (x, y, x, y)$. Since $e_1 = -e_3$ and $e_2 = -e_4$, the first sum vanishes and the second sum partially cancels out, leaving us with

$$\tau_{\nabla} \llbracket x \cdot u, y \cdot v \rrbracket_p = xy \left([\omega_{p_{x,y}^1}(u)]v - [\omega_{p_{x,y}^2}(v)]u \right) - x^2 [\omega_{p_{x,y}^1}(u)]u - y^2 [\omega_{p_{x,y}^1}(v)]v + \mathcal{E}_{x,y}(\mathbf{t}) \quad (3.36)$$

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